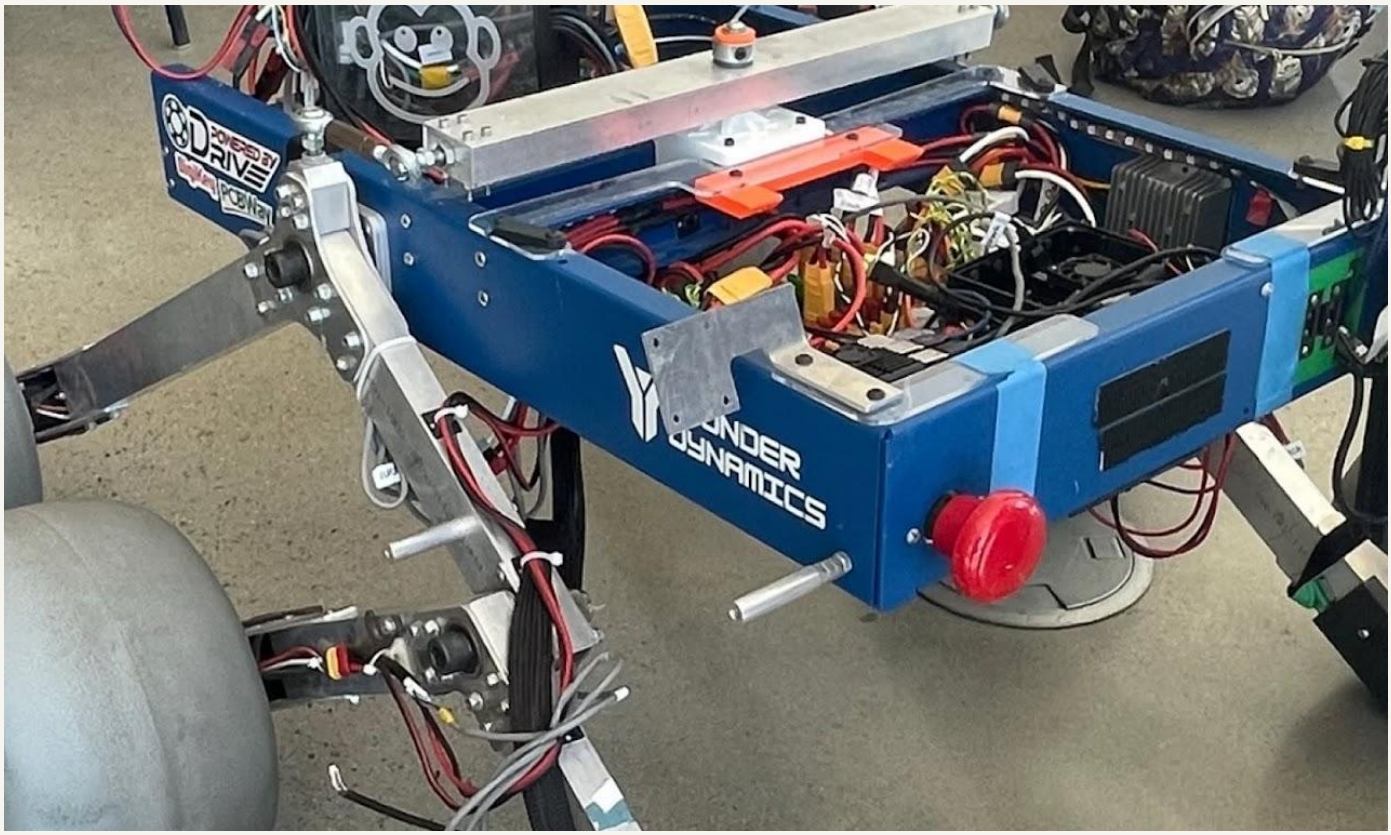
ENGINEERING PORTFOLIO · 2026

Chehak Aggarwal

I design mechanisms for robots and vehicles that work where people can't: Mars-analog terrain, the Arctic ocean, underwater.

SCHOOL	UC San Diego, Jacobs School
DEGREE	B.S. Aerospace Engineering
SPECIALTY	Astrodynamics & Space Apps
GRAD	June 2029 · GPA 3.9
AWARD	Triton Scholarship

CAD	Onshape · CATIA V5 · Fusion 360
ANALYSIS	FEA · MSC ADAMS · kinematics
FAB	3D printing · laser cut · lathe
CODE	Python · MATLAB · GitLab



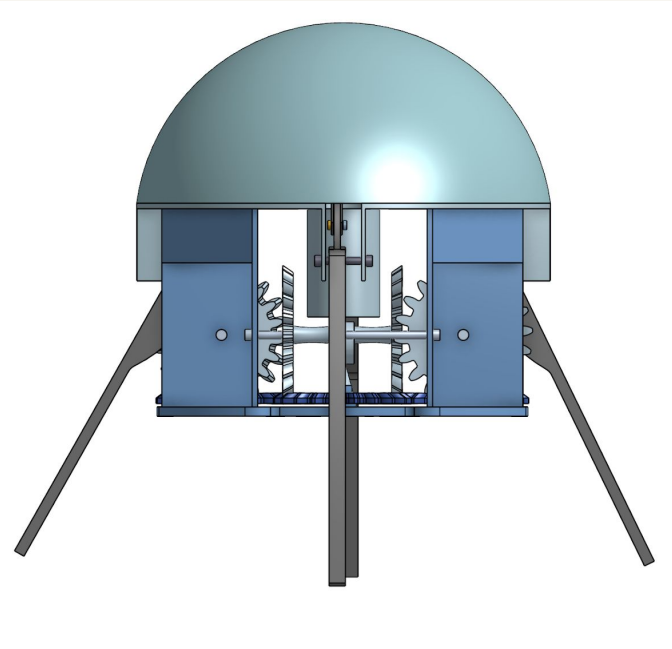
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DWG 04	Lead-screw gripper (adapted) · other work	CAD · motion study

JELIX: jellyfish-inspired underwater propulsion

A propeller-free propulsion mechanism for small underwater vehicles. A cam–follower drive with spring return snaps four ribs inward to push out a jet of water, copying the pulse of Cotylorhiza tuberculata. I designed the cam in CATIA V5 and simulated it in MSC ADAMS, supervised by Theja K V.

65 mm	80 mm	0–40°	12 s
rib-tip travel X	rib-tip travel Y	rib stroke	cycle at 5 rpm



HOW IT WORKS

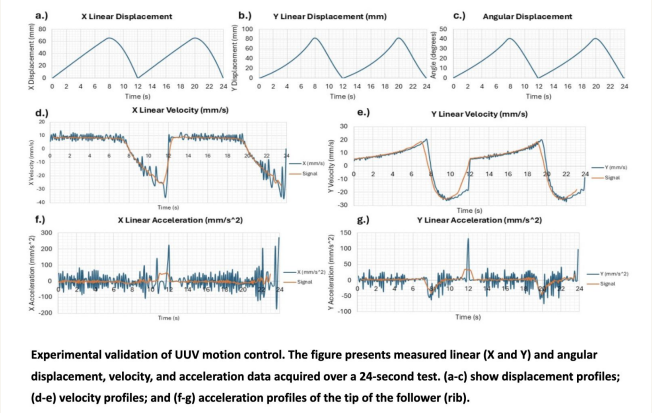
- Expand: the cam slowly raises the ribs and compresses the springs.
- Contract: the cam profile drops, the springs snap the ribs back, and a jet of water pushes the vehicle.

ANALYSIS

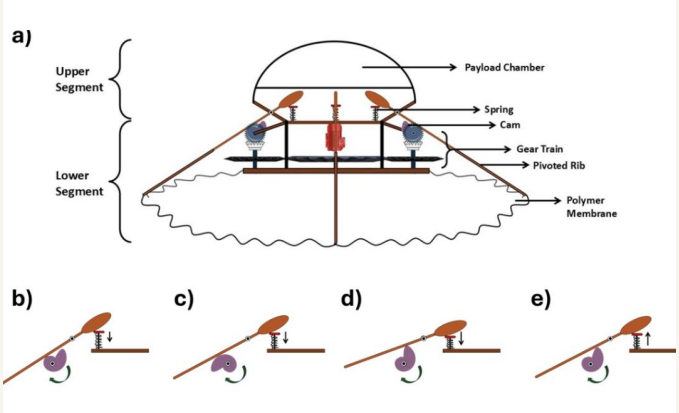
- Cam profile designed at 10° increments in CATIA V5.
- MSC ADAMS multibody simulation at 5 rpm over 24 s.
- Free-body model: Fy on each rib produces thrust.

WHY A CAM

One motor, low voltage, simple to build and maintain. Smart materials need more complex electronics.



Experimental validation of UUV motion control. The figure presents measured linear (X and Y) and angular displacement, velocity, and acceleration data acquired over a 24-second test. (a-c) show displacement profiles; (d-e) velocity profiles; and (f-g) acceleration profiles of the tip of the follower (rib).

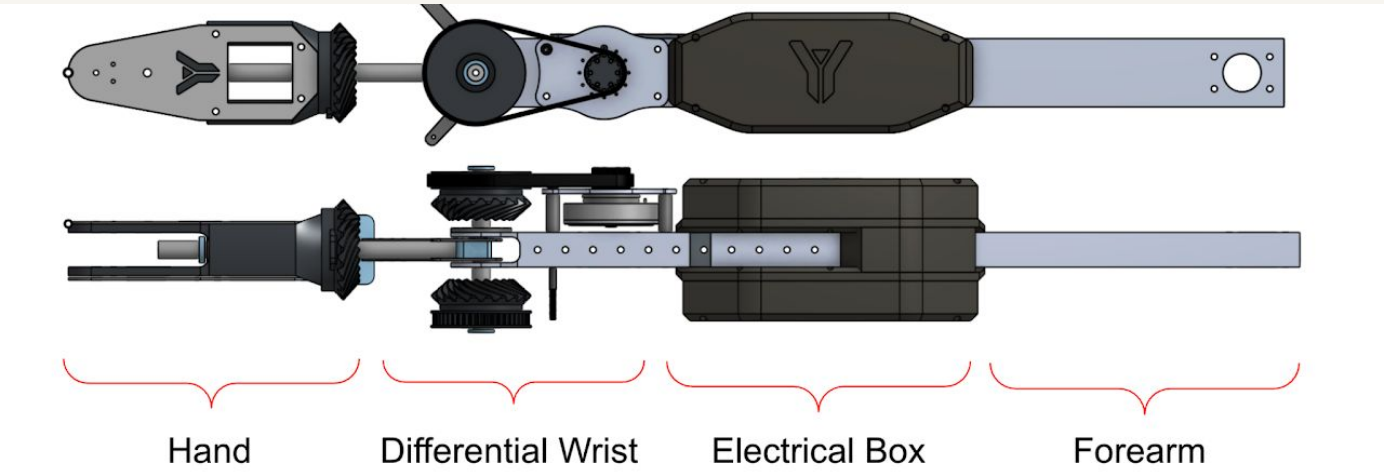


Status: simulation only, not built.

PROJECT	TOOLS / CONTACT	REV	SHEET
JELIX	CATIA V5 · MSC ADAMS	B	2 / 5

DWG 02 · YONDER DYNAMICS (UNIVERSITY ROVER CHALLENGE) · NOV 2025 – PRESENT

Mars rover arm: science claw & enclosure



1 · SCIENCE CLAW: DESIGNED, BUILT, USED

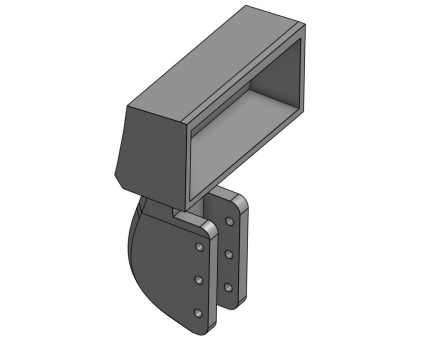
The fingers of the old end effector was only made to grab small objects. I designed a claw to scoop samples and replace fingers and a direct bolt-on interface to the hand.

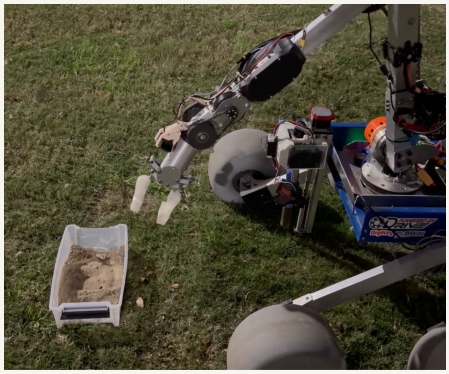
Result: collected geological samples at URC 2026.

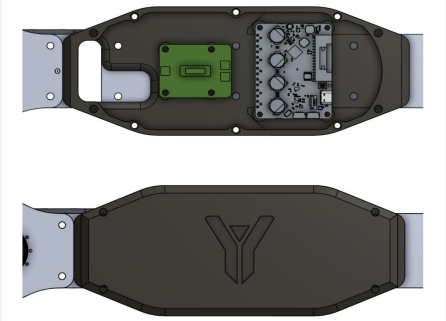
2 · FOREARM ENCLOSURE: REDESIGNED

The new ODrive and fuse board didn't fit; cramped wiring pressed on the fuse board and made servicing hard.

- Length 178 → 216 mm (+21%) · 3 mm walls
- Enlarged cable exit for bend radius · taller cap
- Status: PLA prototype; arm module being redesigned

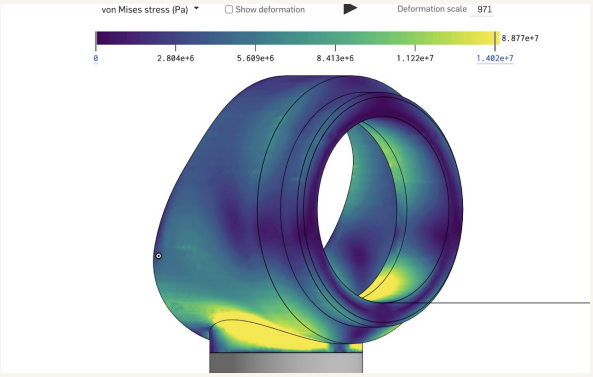






3 · ELBOW FEA: ANALYSIS SUPPORT

Ran static von Mises FEA on a teammate's elbow-joint design and found stress concentrations at the mounting interfaces.

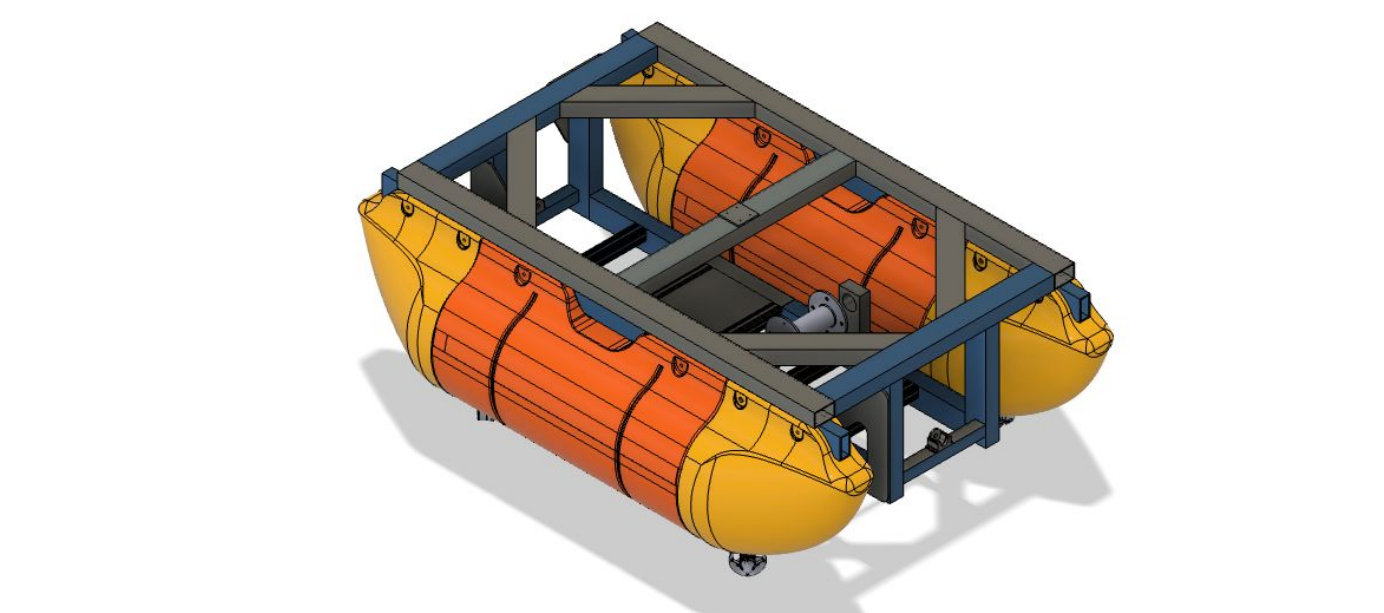


PROJECT	TOOLS / CONTACT	REV	SHEET
ROVER ARM	Onshape · FEA · 3D printing	B	3 / 5

DWG 03 · YONDER DEEP · MECHANICAL TEAM · NOV 2025 – PRESENT

Arctic USV: frame, LiDAR mount, pool tests

An autonomous surface vessel that will collect underwater acoustic data with a hydrophone in the Arctic Ocean. I work on its structure and sensor integration, and helped run its first pool tests.



1 · PONTOON FRAME: CO-DESIGNED

- Material: 6061-T6 aluminum
- Joining: welded + bolted
- Driver: modular, field-serviceable

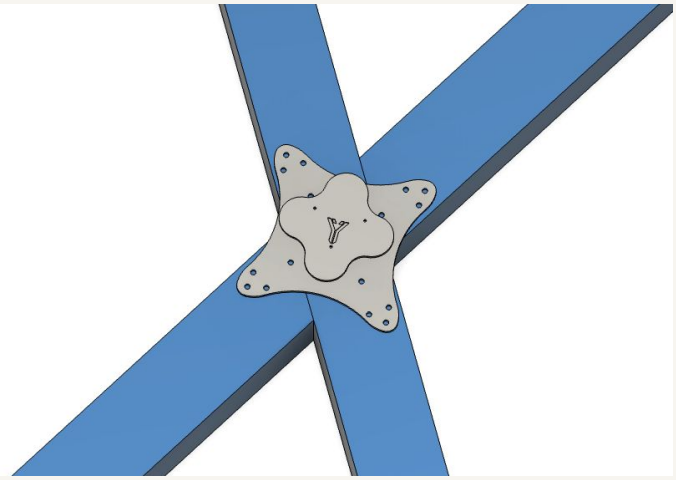
Co-designed with one other mechanical member, through team-wide review rounds.

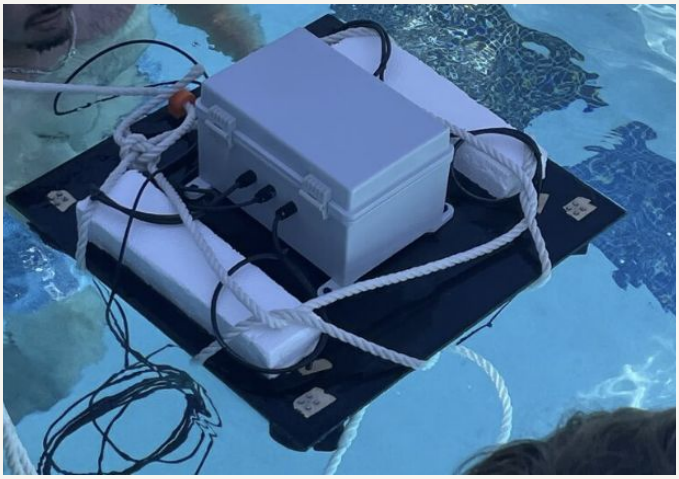
2 · LIDAR MOUNT: DESIGNED

Clean-sheet mount in Fusion 360 for stable sensing and a clear field of view in a marine environment.

3 · POOL TESTS: SUPPORTED

- Preliminary rig, X-configuration thrusters.
- Checked buoyancy, power draw, and hydrophone noise (no interference).
- Test 1: one motor failed. Test 2: out-and-back run completed.

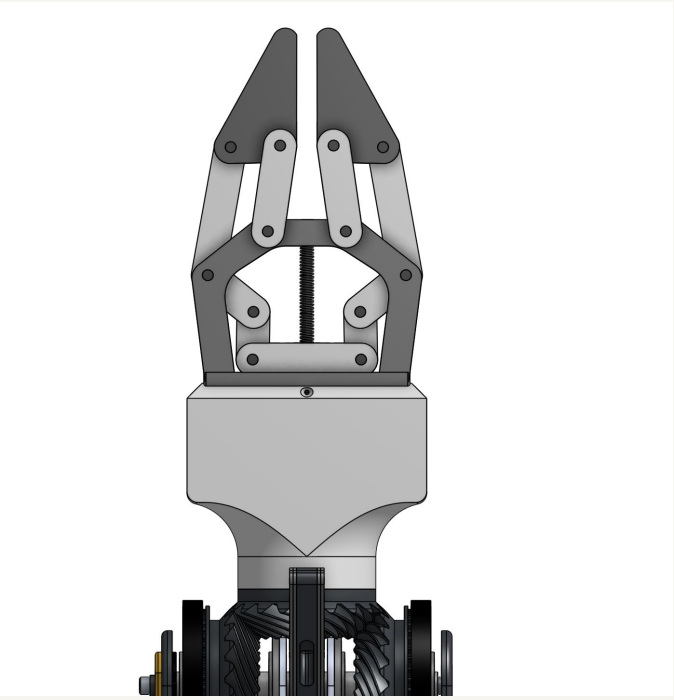




PROJECT	TOOLS / CONTACT	REV	SHEET
ARCTIC USV	Fusion 360 · 3D printing	B	4 / 5

DWG 04 · ADAPTED DESIGN · ONSHAPE · OCT 2026

Lead-screw robotic gripper



A central lead screw drives a carriage that pushes mirrored four-bar linkages, swinging two fingertips together. I adapted an existing design and modeled it with a motion study in Onshape as a Yonder Dynamics design task.

WHY A LEAD SCREW

- High force from a small motor.
- Can be self-locking, so it holds without power.
- Trade-off: slower to close than a direct servo.

Status: CAD and motion study; not manufactured. Motion video on the portfolio site.

Other work

HuLC ’26	NASA Human Lander Challenge: co-researched T-PANS, a tunable viscoelastic damping concept for life-support noise on crewed lunar missions; wrote the Design Rationale & Value section.
AIAA	Project Coordinator: recruited 32 students from 57+ applicants onto four NASA competition teams (HuLC, RASC-AL, Blue Skies, ORBITS).
INTERN	Big Data Rhino: Python analysis of satellite data for proof-of-concept products; taught AI classes to 48 participants.
ALSO	International Aerospace Olympiad, 2nd rank · competitive boxer (South Asian Games gold) · digital artist (64k followers)